method to_java in order to be passed back and forth. See the JRuby[8] documentation for full details.

If you are able to use JRuby, meaning that your code is compliant with Ruby 3.1 and doesn't use gems that use native C code (like Nokogiri), you can get a significant (2x - 3x) performance boost. If you are working in an environment that already has a Java backend, access to the Java libraries and resources can be a big win. (Be careful, though: one of us once worked on a JRuby project that had two User classes that both backed to the same database table, one using Ruby and ActiveRecord, the other using Java and Hibernate. This was not helpful.)

mRuby

mRuby (Minimalist Ruby) is an offshoot of official Ruby, led by many of the same developers. It implements a subset of Ruby, designed for a minimal memory footprint and for use embedded inside C programs where memory might be tight, such as inside devices. The idea is to allow hardware developers access to Ruby as a scripting language.

You can install mRuby via your ruby version manager (rbenv install mruby-3.1.0) or you can install it standalone from the download site.[9] If you install it via a version manager, you can run it using ruby; if you install it standalone, the command is mruby.

Other Runtimes

There have been other attempts to create Ruby runtimes that are either in progress, not used much, or have been abandoned.

Artichoke Ruby

Artichoke Ruby is an attempt to build a CRuby compliant Ruby runtime in Rust. It is currently in pre-production.

Opal

Opal is a Ruby to JavaScript compiler.

MagLev

MagLev is a Ruby runtime built on top of the GemStone Smalltalk runtime. It appears to have had very little development since 2016.

Rubinius

Rubinius is an attempt to build a Ruby runtime in Ruby, partially for use as a reference implementation. It appears to have had little development since 2020.

Iron Ruby

Iron Ruby was a .NET implementation of Ruby, that appears to have had little development since 2011.

8. https://github.com/jruby/jruby/wiki/CallingJavaFromJRuby
9. https://mruby.org/downloads